

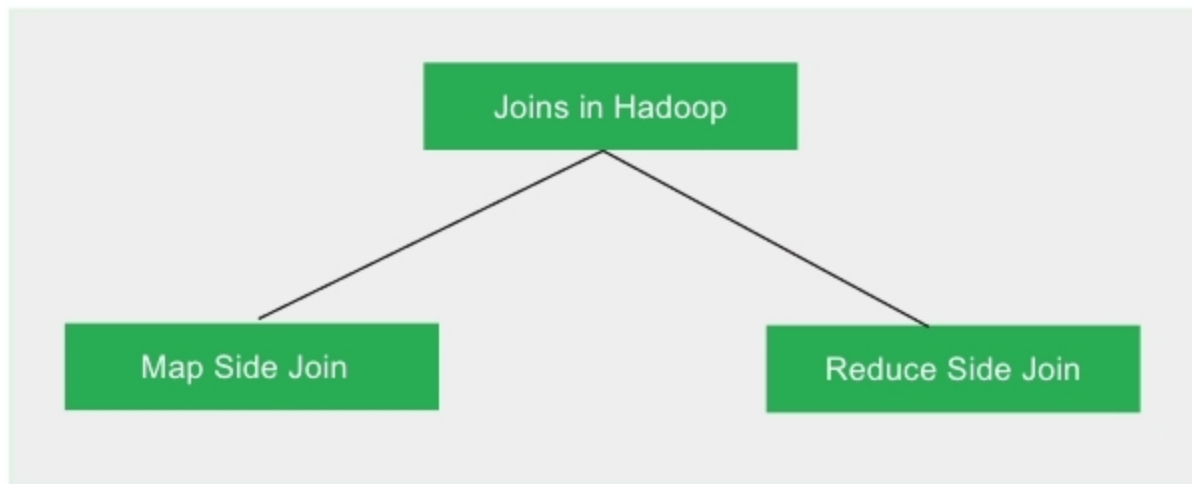


Figure 1: Joins in Hadoop

Difference between map side joins and reduce side joins

1. A map side join, as explained earlier, happens on the map side whereas a reduce side join happens on the reduce side.
2. A map side join happens in the memory whereas a reduce side join happens off the memory.
3. Map side joins are effective when

one data set is big while the other is small, whereas reduce side joins work effectively for big size data sets.

4. Map side joins are expensive, whereas reduce side joins are cheap.

Opt for map side joins when the table size is small and fits in memory, and you require the job to be completed in a short span of time. Use the reduce side join when dealing with large data sets, which cannot fit into the memory. Reduce side joins are easy to implement and have the advantage of their inbuilt sorting and shuffling algorithms. Besides this, there is no requirement to strictly follow any formatting rule for input in case of reduce side joins, and these could also be performed on unstructured data sets. **END** 🐧

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of ICP (individual care process) with the ability to monitor the living conditions using IoT technologies. Thus the Semantic Web of Things (SWOT) and ML algorithms, when combined together, result in LDC (less differentiated caregiver). The resultant integrated healthcare framework can provide significant savings while improving general health.

Machine learning algorithms, techniques and machinery are already present in the market to implement reasonable LDC processes. Thus, this technology is sometimes described as supervised or predictive ML.

IoT in home healthcare systems comprises multi-tier area networks. These consist of body area networks (BAN), the LAN and ultimately the WAN. These also need highly secured hybrid clouds.

IoT devices in home healthcare include nano sensors attached to the skin of the patient's body to measure blood pressure, sugar levels, the heart beat, etc. This raw data is transmitted

to the patient's database that resides in the highly secured cloud platform. The doctor can access the raw data, previous prescriptions, etc, using sophisticated ML algorithms to recommend specific drugs to patients at remote places if required. Thus, patients at home can be saved from life threatening health conditions such as sudden heart attacks, paralysis, etc.

In this era of communication and connectivity, individuals have multiple technologies to support their day-to-day requirements. In this scenario,

IoT together with ML is emerging as a practical solution for problems facing several sectors.

Growth in IoT is fine but just how much of the data collected by IoT devices is actually useful, is the key question. To answer that, efficient data analytics software, open source platforms and cloud technologies should be used. Machine learning and IoT should work towards creating a better technology, which will ensure efficiency and productivity for all sectors. **END** 🐧

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